

Query Results

Grad Café Applicant Data — SQL Analysis Results

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Q1

QUESTION

How many entries are from applicants who applied for Fall 2026? (using the `term` field)

SQL QUERY

```
SELECT COUNT(*) FROM applicants WHERE term ILIKE 'Fall 2026';
```

RESULT

Fall 2026 applicant count: 32,344

EXPLANATION

This SQL query returns a raw count of every entry that has a case-insensitive exact match in the field `term` of `Fall 2026` from the `applicants` table. It's a straightforward count, no missing-value handling needed since `term` is populated for all but one malformed record which was already excluded during loading.

Q2

QUESTION

Among entries that provide a nationality classification, what percentage are international students? (American and Other do not count as international; missing/blank values are excluded from the denominator entirely.)

SQL QUERY

```
SELECT
    SUM(CASE WHEN us_or_international ILIKE 'International' THEN 1 ELSE 0 END),
    COUNT(*)
FROM applicants
WHERE us_or_international IS NOT NULL AND TRIM(us_or_international) <> '';
```

RESULT

Percent international: 48.58% (28558/58786 entries with a usable classification)

EXPLANATION

This SQL query iterates over every entry in the table `applicants`, checks the column `us_or_international` and sums up all entries that have `International` in that column, this sum becomes our numerator. The `ILIKE` syntax tells our query to match regardless of case sensitivity issues. The `WHERE` clause acts as a filter that runs before any counting happens, it removes rows where `us_or_international` is `NULL` outright, and separately removes rows where the value is just whitespace, by trimming the string and checking if the result is empty. Only rows that survive this filter are passed on to be counted: the numerator sums how many of those surviving rows match `'International'`, while the denominator (`COUNT(*)`) simply counts all of the surviving rows regardless of their value, which is what makes the resulting percentage meaningful.

Q3

QUESTION

What are the average GPA, GRE (Quantitative), GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric? (each average computed independently)

SQL QUERY

```
SELECT AVG(gpa), COUNT(gpa) FROM applicants WHERE gpa IS NOT NULL;  
SELECT AVG(gre), COUNT(gre) FROM applicants WHERE gre BETWEEN 130 AND 170;  
SELECT AVG(gre_v), COUNT(gre_v) FROM applicants WHERE gre_v BETWEEN 130 AND 170;  
SELECT AVG(gre_aw), COUNT(gre_aw) FROM applicants WHERE gre_aw BETWEEN 0 AND 6;
```

RESULT

- Average GPA: 3.77 (n=34645)
- Average GRE: 165.87 (n=1818)
- Average GRE V: 160.44 (n=3830)
- Average GRE AW: 4.33 (n=3428)

EXPLANATION

Disclaimer: A simpler query was first run to calculate these four averages directly from the raw data. However, the GRE Quantitative average came back far higher than the highest possible score on the current scale, which revealed that the gre, gre_v, and gre_aw columns contain contaminated data. The likely cause is misplaced values like combined scores and sentinel/placeholder values. Because of this, each GRE-related query below was modified to filter out invalid values before averaging.

Each line of SQL corresponds to each of the four questions. The first one selects all rows where gpa is not null. The second line selects all rows where gre is not null AND falls between 130 and 170. The third line selects all rows where gre_v is not null AND falls between 130 and 170. The fourth line selects all rows where gre_aw is not null AND falls between 0 and 6. All four queries then calculate the average of those scores, and returns the raw count.

Q4

QUESTION

What is the average GPA of American applicants who applied for Fall 2026?

SQL QUERY

```
SELECT AVG(gpa), COUNT(gpa) FROM applicants
WHERE term ILIKE 'Fall 2026'
      AND us_or_international ILIKE 'American'
      AND gpa IS NOT NULL;
```

RESULT

Average GPA, American applicants, Fall 2026: 3.79 (n=11396)

EXPLANATION

The WHERE filters to entries that are simultaneously Fall 2026, classified as American, and have a non-null GPA. The query then calculates the average GPA of those entries, and returns the raw count value.

Q5

QUESTION

What percentage of Fall 2025 entries are acceptances?

SQL QUERY

```
SELECT
    SUM(CASE WHEN status ILIKE 'Accepted' THEN 1 ELSE 0 END),
    COUNT(*)
FROM applicants
WHERE term ILIKE 'Fall 2025';
```

RESULT

Fall 2025 acceptance percentage: 39.30% (10641/27074 entries)

EXPLANATION

The WHERE filters to only entries that have `Fall 2025` in the `term` column from our table applicants. The denominator is the count of all Fall 2025 entries regardless of status and the numerator is the subset which is counted by summing every entry that case insensitive matches with `Accepted` in the `status` column.

Q6

QUESTION

What is the average GPA of accepted applicants who applied for Fall 2026?

SQL QUERY

```
SELECT AVG(gpa), COUNT(gpa) FROM applicants
WHERE term ILIKE 'Fall 2026'
      AND status ILIKE 'Accepted'
      AND gpa IS NOT NULL;
```

RESULT

Average GPA, accepted applicants, Fall 2026: 3.78 (n=7696)

EXPLANATION

We're selecting from the applicants table. The WHERE clause filters entries that have a GPA value, in the `status` column have none case sensitive match of `Accepted` and none case sensitive `Fall 2026` in the `term` column. The query then calculates the average GPA of those entries, and returns the raw count.

Q7

QUESTION

How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science? Use the original downloaded fields, particularly: program, degree. Matching should reasonably recognize: Johns Hopkins University / JHU; Computer Science; and master's degree entries.

SQL QUERY

```
-- from module_3/query_data.py
JHU_PATTERN = r"\yJohns? Hopkins\y" -- accepts the "John Hopkins" typo
CS_PATTERN = r"\y(Computer Sciences?|CS)\y"

SELECT COUNT(*) FROM applicants
WHERE program ~* '\yJohns? Hopkins\y'
      AND program ~* '\y(Computer Sciences?|CS)\y'
      AND degree = 'Masters';
```

RESULT

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EXPLANATION

This query checks the program field for two things at once, using regex pattern matching. The first condition is whether it mentions Johns Hopkins (including the common misspelling 'John Hopkins,'), and second, whether it mentions Computer Science. Computer Science can either be written out fully or as the abbreviation 'CS.' Both checks use word-boundary anchors (\y in Postgres regex) so that 'CS' only matches when it appears as its own standalone word, not as a substring buried inside unrelated words like 'Economics' or 'Physics.' The query also restricts results to degree = 'Masters', since the question specifically asks about master's-level applicants. Then it simply counts the total number of entries that match these constraints.

Q8

QUESTION

Using the original downloaded fields - How many Fall 2026 entries are acceptances from applicants applying for a PhD in Computer Science at one of the following universities? Georgetown University Massachusetts Institute of Technology / MIT Stanford University Carnegie Mellon University The query simultaneously restricts by: Fall 2026 term acceptance status PhD degree Computer Science and one of the four universities listed

SQL QUERY

```
-- from module_3/query_data.py
CS_PATTERN = r"\y(Computer Sciences?|CS)\y"
GEORGETOWN_PATTERN = r"Georgetown University"      -- literal, excludes
"Georgetown College" (KY)
MIT_PATTERN = r"\y(MIT|Massachusetts Institute of Technology)\y"
STANFORD_PATTERN = r"\yStanford\y"
CARNEGIE_MELLON_PATTERN = r"\yCarnegie Mellon\y"

SELECT COUNT(*) FROM applicants
WHERE term = 'Fall 2026'
      AND status = 'Accepted'
      AND degree = 'PhD'
      AND program ~* '\y(Computer Sciences?|CS)\y'
      AND (
          program ~* 'Georgetown University'
        OR program ~* '\y(MIT|Massachusetts Institute of Technology)\y'
        OR program ~* '\yStanford\y'
        OR program ~* '\yCarnegie Mellon\y'
      );
```

RESULT

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EXPLANATION

One combined count across all four target universities (not per-university), all filters applied simultaneously against the original `program / term / status / degree` fields. Per-university breakdown under these same filters: Georgetown=0, MIT=19, Stanford=7, Carnegie Mellon=4.

Q9

QUESTION

Repeat Question 8, but identify the university and program using `llm_generated_program` and `llm_generated_university` instead of the original downloaded university/program information. Continue using the original fields for: term; degree; and admission status. Report the Q8 count, the Q9 count, and the difference between them.

SQL QUERY

```
-- from module_3/query_data.py
SELECT COUNT(*) FROM applicants
WHERE term = 'Fall 2026'
      AND status = 'Accepted'
      AND degree = 'PhD'
      AND llm_generated_program ~* '\y(Computer Sciences?|CS)\y'
      AND (
          llm_generated_university ~* 'Georgetown University'
        OR llm_generated_university ~* '\y(MIT|Massachusetts Institute of
Technology)\y'
        OR llm_generated_university ~* '\yStanford\y'
        OR llm_generated_university ~* '\yCarnegie Mellon\y'
      );
```

RESULT

Q8 count: 30 Q9 count: 26 Difference (Q8 – Q9): 4

EXPLANATION

Q8 and Q9 apply identical filters (Fall 2026, Accepted, PhD, Computer Science, one of four target universities) and differ only in which fields are used to identify the university and program. Q8 uses the original scraped `program` field and Q9 uses `llm_generated_university` and `llm_generated_program`, produced by a small local LLM.

The counts differ by 4, and the entire gap is attributable to a single failure in the LLM standardization step. For every applicant who listed "Carnegie Mellon University" in the original data, the LLM output "Carnegie Melon University" in `llm_generated_university`. Because our matching pattern for Carnegie Mellon (`\yCarnegie Mellon\y`) requires the correct spelling, all 4 Carnegie Mellon PhD/CS/Fall 2026 acceptances that were correctly counted in Q8 are silently dropped from Q9.

Custom Question 1

QUESTION

What percentage of GRE Quantitative score submissions in the database are contaminated with misplaced-value or sentinel-style entries (i.e., fall outside the plausible 130–170 current-scale range)?

SQL QUERY

```
SELECT
    SUM(CASE WHEN gre NOT BETWEEN 130 AND 170 THEN 1 ELSE 0 END),
    COUNT(*)
FROM applicants
WHERE gre IS NOT NULL;
```

RESULT

Custom Q1 (GRE Quant contamination rate): 60.99% (2842/4660 entries outside the plausible 130-170 range)

EXPLANATION

This question builds off of the data quality issues from Q3. It applies the inverse filter, rather than selecting only valid GRE scores, it selects all of the GRE scores that do not fall in the valid range, and then calculates a percentage to give us an idea of how widespread the issue is. There are 4660 non-null GRE Quant entries, and 2842 (61%) fall outside the plausible current-scale range [130,170]. 42 below 130, 2800 above 170.

Custom Question 2

QUESTION

What is the acceptance rate across all degree types represented in the dataset, and how does it compare between PhD and Master's applicants specifically?

SQL QUERY

```
SELECT
    degree,
    SUM(CASE WHEN status ILIKE 'Accepted' THEN 1 ELSE 0 END) AS accepted,
    COUNT(*) AS total
FROM applicants
WHERE degree IS NOT NULL
GROUP BY degree
ORDER BY total DESC;
```

RESULT

Degree	Accepted	Total	Acceptance rate
PhD	10197	40587	25.12%
Masters	11368	16880	67.35%
MFA	450	1311	34.32%
PsyD	247	670	36.87%
Other	282	458	61.57%
EdD	29	45	64.44%
MBA	33	43	76.74%
JD	15	30	50.00%

EXPLANATION

Masters applicants are accepted at roughly 2.7x the rate of PhD applicants, 67.35% vs. 25.12%. This is a large gap and plausibly reflects real world differences between the two admissions processes. PhD programs are typically far more selective, fund every admit, and have many more applicants per seat, while Masters programs are often larger, less funding-constrained, and more numbers-driven in admissions. MBA (76.74%) and EdD (64.44%) show similarly high acceptance rates to Masters. MFA (34.32%) and PsyD (36.87%) sit closer to PhD's selectivity, however though these smaller-n categories should be read with more caution than the PhD/Masters comparison.